Lab 3

Submission by: **March 24th - 4pm**

1) [0.20 points]
   Give regular expressions generating the following languages over the alphabet $\Sigma = \{0,1\}$:
   - **a.** $\{w \mid w$ begins with a 1 and ends with a 0$\}$
   - **b.** $\{w \mid w$ contains at least three 1s$\}$
   - **c.** $\{w \mid w$ contains the substring 0101 (i.e., $w = x0101y$ for some x and y)$\}$
   - **d.** $\{w \mid w$ has length at least 3 and its third symbol is a 0$\}$
   - **e.** $\{w \mid w$ starts with 0 and has odd length, or starts with 1 and has even length$\}$
   - **f.** $\{w \mid w$ doesn't contain the substring 110$\}$
   - **g.** $\{w \mid$ the length of w is at most 5$\}$
   - **h.** $\{w \mid w$ is any string except 11 and 111$\}$
   - **i.** $\{w \mid$ every odd position of w is a 1$\}$
   - **j.** $\{w \mid w$ contains at least two 0s and at most one 1$\}$
   - **k.** $\{\varepsilon, 0\}$
   - **l.** $\{w \mid w$ contains an even number of 0s, or contains exactly two 1s$\}$
   - **m.** The empty set
   - **n.** All strings except the empty string

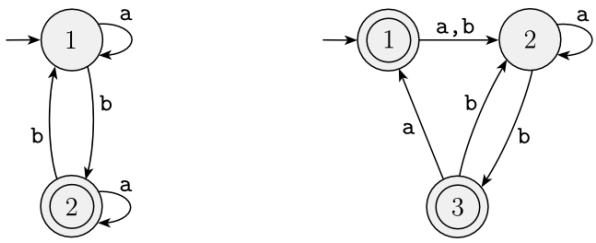
2) [0.30 points]

Use the construction proof to convert the following regular languages to NFAs:

a. $(0 \cup 1)^* 010 (0 \cup 1)^*$
b. $(((11)^*(00)) \cup 01)^*$
c. $\emptyset^*$

3) [0.40 points]

Use the construction proof to convert the following finite automata to regular expressions:



(a) (b)

4) [0.60 points]

In certain programming languages, comments appear between delimiters such as `/#` and `#/`. Let C be the language of all valid delimited comment strings. A member of C must begin with `/#` and end with `#/` but have no intervening `#/`. For simplicity, assume that the alphabet for C is $\Sigma = \{\mathtt{a}, \mathtt{b}, /, \#\}$.

a. Give a DFA that recognizes C.

b. Give a regular expression that generates C.

5) [0.5 points]

For any string $w = w_1 w_2 \cdots w_n$, the ***reverse*** of w, written $w^{\mathcal{R}}$, is the string w in reverse order, $w_n \cdots w_2 w_1$. For any language A, let $A^{\mathcal{R}} = \{w^{\mathcal{R}} | \, w \in A\}$. Show that if A is regular, so is $A^{\mathcal{R}}$.

6) [0.5 points]

Let $B_n = \{\mathtt{a}^k | \, k \text{ is a multiple of } n\}$. Show that for each $n \geq 1$, the language B_n is regular.